FRE-GY 9743 Assignment 4 (Math Part)

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Q1. Discounting: OTC vs. Exchange-Traded Futures

Question. Both OTC and exchange-traded trades involve daily collateral/margin payments of "MTM − existing collateral". Why does the OTC price carry a discount factor while the futures price does not?

Short answer. The daily cash transfers look the same but mean different things. In a futures contract the daily payment is the *settlement* of the contract: the gain is paid out for good and the contract restarts at value 0. In an OTC trade the collateral is only a *deposit* that secures the trade: it is returned with interest, and the contract is still alive and still owes its payoff at T. So a futures contract has only one day of value left to discount (and that factor cancels), while an OTC contract has the whole remaining life to discount.

1) Futures: the daily payment is final. At t_{k+1} we receive $F(t_{k+1}, T) - F(t_k, T)$, and the contract is then worth 0 again. At t_k, the only thing we are owed is this one payment, one day away, so

$$0 \;=\; D(t_k, t_{k+1})\, \mathbb{E}^{\mathbb{Q}}[F(t_{k+1}, T) - F(t_k, T) \,|\, \mathcal{F}_{t_k}]\,.$$

Here $D(t_k, t_{k+1})$ is known at t_k and positive, and it multiplies an expression whose value is 0. Scaling a zero by a positive number changes nothing, so the discount factor drops out and $F(t_k, T) = \mathbb{E}^{\mathbb{Q}}[F(t_{k+1}, T) \,|\, \mathcal{F}_{t_k}]$. Iterating to T gives

$$F(t_k, T) = \mathbb{E}^{\mathbb{Q}}[S(T) \,|\, \mathcal{F}_{t_k}].$$

No $D(t, T)$ ever appears, because nothing is left to be paid at T: all the value has already been paid out along the way.

2) OTC: collateral is a deposit, not a settlement. Take a forward struck at K paying $S(T) - K$ at T, with mark-to-market $V(t)$. The collateral held at t_k is $C(t_k) = V(t_k)$, and on each date we post $V(t_{k+1}) - C(t_k)$. But collateral stays our money: the holder pays interest on it at the collateral rate (the discount rate) and returns it when the trade ends. Depositing C at t_k and getting back $C/D(t_k, t_{k+1})$ at t_{k+1} has present value

$$D(t_k, t_{k+1}) \cdot \frac{C}{D(t_k, t_{k+1})} - C = 0,$$

so posting collateral creates no value. The only real cash flow of the trade is $S(T) - K$ paid at T, and it must be discounted over the full remaining life:

$$V(t_k) = \mathbb{E}^{\mathbb{Q}}\big[D(t_k, T)\,\big(S(T) - K\big)\,\big|\,\mathcal{F}_{t_k}\big]\,.$$

Equivalently, $V(t_k) \;=\; D(t_k, t_{k+1})\, \mathbb{E}^{\mathbb{Q}}[V(t_{k+1}) \,|\, \mathcal{F}_{t_k}]$. This is the same one-day argument as for futures, but now the contract is *not* reset to 0 after the margin call (the right-hand side is $V(t_{k+1}) \neq 0$), so the factor does not cancel. Iterating to T accumulates $D(t_k, T)$.

3) Intuition. Think of a futures contract as paying rent each day: the payment is final and the lease restarts, so only today's payment matters. An OTC trade is like a security deposit: you get it back with interest, and the real payment still sits at T, so it is worth less today and must be discounted.

	Futures (exchange)	**OTC with collateral**
Daily payment is	settlement of the P&L	a deposit, returned with interest
Value after payment	0 (contract resets)	$V \neq 0$ (contract still alive)
Cash flow left at T	none	$S(T) - K$
Discounting	one day, cancels	$D(t,T)$, stays
Result	$F = \mathbb{E}^{\mathbb{Q}}[S(T)]$	$K = \dfrac{\mathbb{E}^{\mathbb{Q}}[D(t,T)S(T)]}{\mathbb{E}^{\mathbb{Q}}[D(t,T)]}$

4) Consequence. The forward price satisfies

$$K - F = \frac{\mathrm{Cov}^{\mathbb{Q}}\big(D(t,T), S(T)\big)}{\mathbb{E}^{\mathbb{Q}}[D(t,T)]}.$$

With deterministic rates the covariance is zero and $F = K$. With stochastic rates they differ by a convexity adjustment: if S is positively correlated with rates, the futures gains are reinvested at higher rates and the losses financed at lower rates, so $F > K$.